

Anova

① Null ② Alt $\rightarrow$ testz-test $\rightarrow$ t-test
chi-sq

F-test

 x_1, x_2, x_3
1, 1, 1

Hypothesis testing in Anova

Null hypothesis (H_0) $\Rightarrow \mu_1 = \mu_2 = \mu_3 = \dots = \mu_n$ Alternate hypothesis $\Rightarrow (H_1) \Rightarrow$ At least one of the sample is not equal $\mu_1 \neq \mu_2 \neq \mu_3 = \dots = \mu_k$ 

(WOODR)

- ① Doctors want to test a new medication which reduces headache. They split the participants into 3 conditions [15mg, 30mg, 45mg] later on the doctor ask the patient to rate the headache return rate [1-10].

Are there any difference between the 3 condition using $\alpha = 0.05$? $\alpha = 0.05$

15mg	30mg	45mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

① Null & Alternate hypothesis

Null hypothesis (H_0) $= \mu_{15} = \mu_{30} = \mu_{45}$ Alternate hypothesis (H_1) $= \mu_{15} \neq \mu_{30} = \mu_{45}$ not all μ are equal.② Significant val $\alpha = 0.05$ C.I = 0.95

③ Calculate Degree of free

 $N = 21$ $a = 3$ $n = 7$ $df_{\text{error}} = a - 1 \Rightarrow 3 - 1 = 2$

$$df_{\text{between}} = a - 1 \Rightarrow 3 - 1 \Rightarrow 2$$

$$df_{\text{within}} = N - a \Rightarrow 21 - 3 \Rightarrow 18$$

$$df_{\text{total}} = N - 1 \Rightarrow 20$$

(df_1, df_2)

(2, 18)

↓

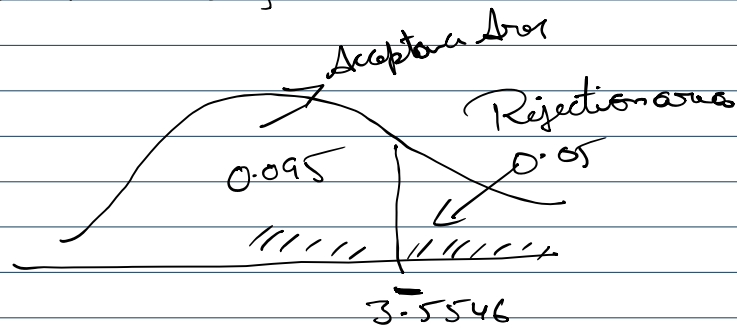
C.V

F_{table}

↓

3.5546

(4) Decision Boundary



Decision Rule

If F_u greater than 3.5546, reject null hypothesis

(5) Calculate F test stat

$$F = \frac{\text{Variance b/w sample}}{\text{Variance with sample}}$$

	SS	df	MS	F
Between	98.62	2	49.31	
within	10.29	18	0.57	
total	108.95	20	5.38	

$$\textcircled{1} SS = \frac{\sum (\sum a_i)^2}{n} - \frac{T^2}{N}$$

SS between

$$15mg \Rightarrow 9+8+7+8+8+9+8 = 57$$

$$30mg \Rightarrow 7+6+6+7+8+7+6 = 47$$

$$45mg \Rightarrow 4+3+2+3+4+3+2 = 21$$

15mg	30mg	45mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2
Σ	Σ	Σ

$$= \frac{57^2 + 47^2 + 21^2}{7} - \frac{57^2 + 47^2 + 21^2}{21}$$

$$= 98.67$$

$$= \frac{\sum y^2 - \frac{(\sum a_i)^2}{n}}{n} = 9^2 + 8^2 + 7^2 + 8 + \dots$$

$$= 853$$

$$= 853 - \frac{(57^2 + 47^2 + 21^2)}{7}$$

$$= 10.29$$

$$F = \frac{M_{S \text{ with } \omega}}{m_{S \omega}} = F = \frac{49.34}{0.54} \Rightarrow 26.56$$

F is greater than 3.5546 value

Reject null hypothesis

$\mu_1 = \mu_2 = \mu_3$ (H_0) $\rightarrow$ ~~Are~~ do not equal ✓

Conclusion.